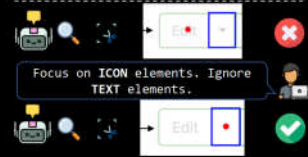
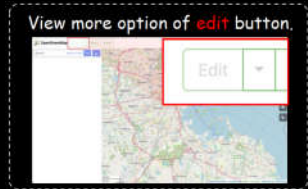
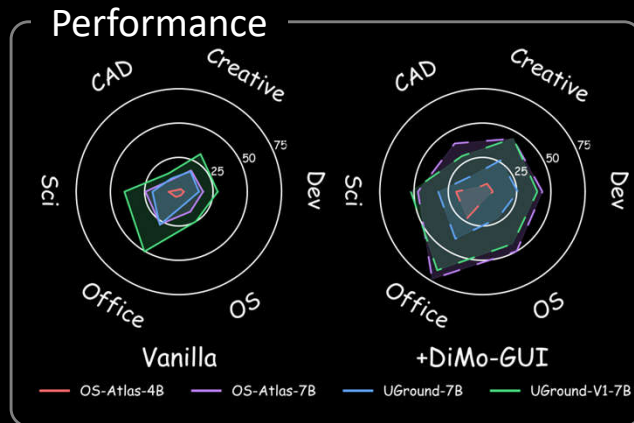
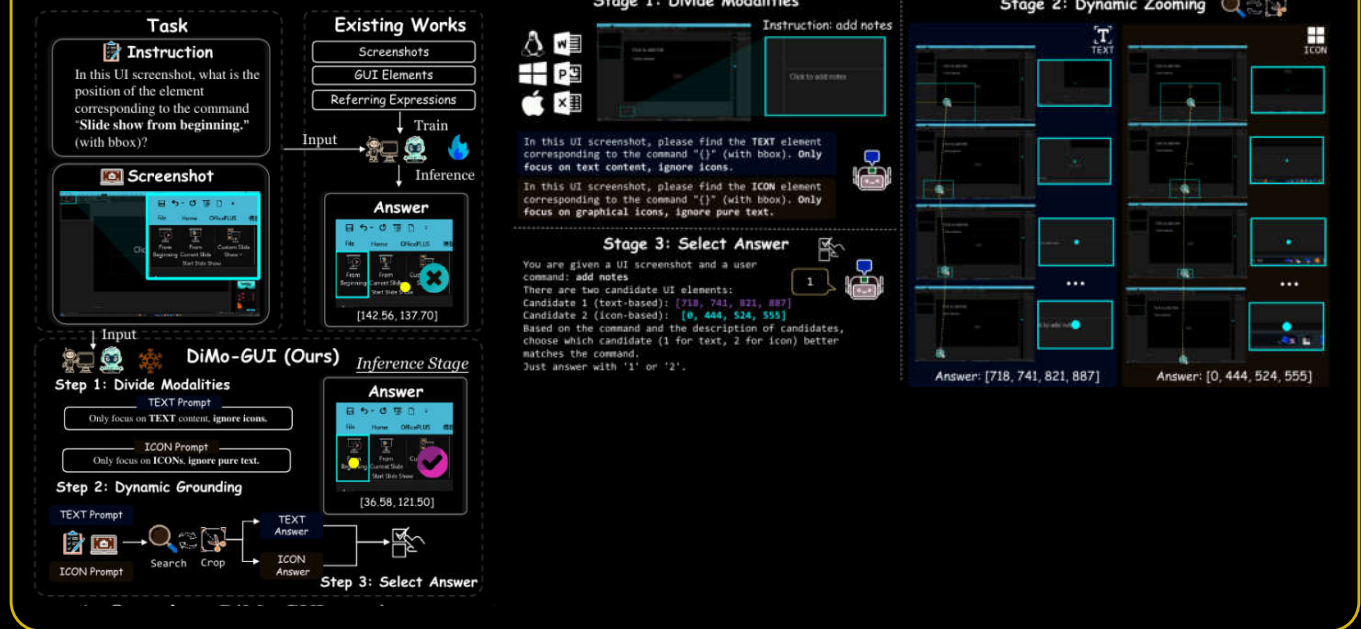
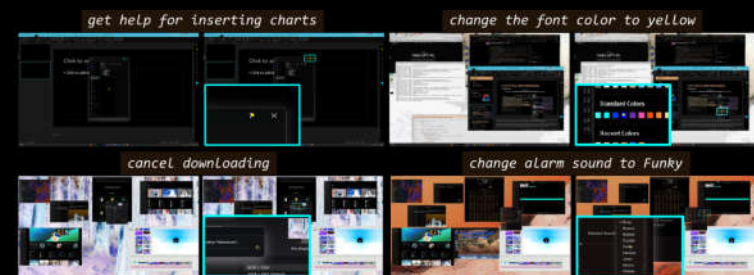


## DiMo-GUI: Test-Time Scaling Gui Grounding



| Method            | OS-Atlas-7B |
|-------------------|-------------|
| vanilla           | 18.4        |
| w DG              | 45.7        |
| w MD              | 26.1        |
| <b>w DiMo-GUI</b> | <b>49.7</b> |



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**Algorithm 1:** Dual-Modality Grounding with Dynamic Zooming

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**Input:** Full-resolution GUI image  $I$ ,  
instruction  $Q$

**Output:** Final grounded coordinate  $C^*$

- ```

1 Step 1: Text modality grounding.
2  $C_{\text{text}} \leftarrow \text{DynamicGrounding}(I, Q, \text{"text"})$ 
3 Step 2: Icon modality grounding.
4  $C_{\text{icon}} \leftarrow \text{DynamicGrounding}(I, Q, \text{"icon"})$ 
5 Step 3: Candidate selection.
6  $C^* \leftarrow \text{Select}(C_{\text{text}}, C_{\text{icon}}, I, Q)$ 
7 return  $C^*$ 

8 Function:  $\text{DynamicGrounding}(I, Q,$ 
    modality  $m)$ 
9 Initialize zoom region:  $R \leftarrow I$ 
10 for  $t = 1$  to  $\text{max\_iters}$  do
11     Predict coordinate:
12      $C_t \leftarrow \text{PredictCoordinate}(R, Q, m)$ 
13     if  $\text{StopCondition}(C_t, t)$  then
14          $\text{return } C_t$ 
15      $R \leftarrow \text{CropAround}(R, C_t)$  // update
        region
16 return  $C_{\text{max}}$ 

```

| GUI Agent MILMs                   | Mobile      |             | Desktop     |             | Web         |             | Average     |
|-----------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                                   | Test        | Icon/Widget | Test        | Icon/Widget | Test        | Icon/Widget |             |
| InternVL-2-BB (Chen et al., 2024) | 9.2         | 4.3         | 4.6         | 4.3         | 0.9         | 0.1         | 4.3         |
| Fuyu (Hlavicek et al., 2023)      | 4.0         | 1.3         | 33.0        | 3.6         | 33.9        | 4.4         | 19.5        |
| Qwen2-VL-7B (Wang et al., 2024b)  | 61.3        | 39.3        | 52.0        | 45.0        | 33.0        | 21.8        | 42.9        |
| CogAgent (Hong et al., 2024)      | 67.0        | 24.0        | 74.2        | 20.0        | 70.4        | 28.6        | 47.4        |
| SeeClick (Cheng et al., 2024)     | 78.0        | 52.0        | 72.2        | 30.0        | 55.7        | 32.5        | 53.4        |
| OS-Atlas-4B (Wu et al., 2024)     | 85.7        | 58.5        | 72.2        | 45.7        | 82.6        | 63.1        | 73.4        |
| UGround-7B (Gou et al., 2024)     | 82.8        | 60.3        | 82.5        | 63.6        | 80.4        | 70.4        | 73.3        |
| OS-Atlas-7B (Gou et al., 2024)    | 92.0        | 72.0        | 91.8        | 69.0        | 89.9        | 74.9        | 82.5        |
| +DMM-GUI                          | <b>96.2</b> | <b>73.8</b> | <b>96.4</b> | <b>75.1</b> | <b>89.7</b> | <b>75.4</b> | <b>85.7</b> |
| UGround-7B (Gou et al., 2024)     | <b>95.0</b> | <b>83.3</b> | <b>95.0</b> | <b>77.8</b> | <b>92.1</b> | <b>77.2</b> | <b>87.6</b> |
| +DMM-GUI                          | <b>94.8</b> | <b>85.3</b> | <b>94.0</b> | <b>82.1</b> | <b>93.2</b> | <b>80.3</b> | <b>89.2</b> |

| Grounding Model                         | GloVeEmbed |      |      | Creative   |      |      | CAD        |      |      | Scientific |      |      | Office     |      |      | DS         |      |      | Avg        |      |      |
|-----------------------------------------|------------|------|------|------------|------|------|------------|------|------|------------|------|------|------------|------|------|------------|------|------|------------|------|------|
|                                         | test score | avg  | std  | test score | avg  | std  | test score | avg  | std  | test score | avg  | std  | test score | avg  | std  | test score | avg  | std  | test score | avg  | std  |
| Qwen1.5-72B (Hu et al., 2023)           | 0.1        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.1  | 0.1  |
| OPT-6.7 (OpenAI, 2022)                  | 0.3        | 0.1  | 0.0  | 1.0        | 0.0  | 0.6  | 1.0        | 0.0  | 1.5  | 2.1        | 0.0  | 1.2  | 1.1        | 0.0  | 0.6  | 0.0        | 0.0  | 0.0  | 1.3        | 0.0  | 1.0  |
| SeaChat (Chen et al., 2024)             | 0.6        | 0.0  | 0.3  | 1.0        | 0.0  | 0.6  | 2.5        | 0.0  | 1.9  | 1.5        | 0.0  | 2.0  | 1.1        | 0.0  | 0.2  | 2.8        | 0.0  | 1.5  | 1.8        | 0.0  | 1.0  |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  |
| MiniCPM3-72B (1.0) (Zhang et al., 2024) | 1.6        | 0.4  | 0.4  | 1.0        | 0.0  | 0.5  | 2.5        | 0.0  | 1.9  | 13.2       | 7.3  | 10.0 | 15.3       | 7.3  | 13.3 | 10.1       | 2.2  | 6.0  | 10.8       | 26.7 | 7.7  |
| Qwen2.5-188B (Huang et al., 2024)       | 1.9        | 0.7  | 0.6  | 0.0        | 0.0  | 0.6  | 5.6        | 7.1  | 3.1  | 61.7       | 22.2 | 13.4 | 114.0      | 0.0  | 6.3  | 5.5        | 0.0  | 3.1  | 120.0      | 0.7  | 3.1  |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  |
| Classic GPT-3.5 (Wei Huo et al., 2024)  | 22.0       | 5.9  | 12.6 | 25.9       | 4.4  | 16.8 | 14.5       | 3.3  | 16.1 | 19.8       | 15.3 | 23.9 | 30.1       | 26.2 | 11.0 | 4.5        | 8.1  | 25.4 | 7.1        | 13.1 | 0.1  |
| US-TARS-70 (Gao et al., 2025)           | 24.4       | 5.6  | 50.1 | 31.2       | 28.8 | 20.4 | 34.4       | 18.0 | 63.9 | 31.5       | 50.0 | 63.3 | 20.8       | 53.5 | 20.8 | 53.5       | 20.8 | 53.5 | 20.8       | 53.5 | 20.8 |
| US-TARS-720k (Gao et al., 2025)         | 63.0       | 49.0 | 49.0 | 77.0       | 8.4  | 39.6 | 1.8        | 12.5 | 16.7 | 16.7       | 16.7 | 63.3 | 54.0       | 54.0 | 54.0 | 54.0       | 18.7 | 30.1 | 40.7       | 18.7 | 30.1 |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  | 0.0        | 0.0  | 0.0  |
| Qwen2.5-72B (Yang et al., 2024)         | 0.3        |      |      |            |      |      |            |      |      |            |      |      |            |      |      |            |      |      |            |      |      |